

Free fatty acids and skeletal muscle insulin resistance.

PURPOSE OF REVIEW: Acute exposure to fatty acids causes insulin resistance in muscle, and excess dietary lipid and obesity are also strongly associated with muscle insulin resistance. Relevant mechanisms, however, are still not fully elucidated. Here we examine the latest evidence as to why lipids might accumulate in muscle and the possible mechanisms for lipid-induced insulin resistance. RECENT FINDINGS: Muscle lipid metabolites such as long chain fatty acid coenzyme As, diacylglycerol and ceramides may impair insulin signalling directly. Crosstalk between inflammatory signalling pathways and insulin signalling pathways, mitochondrial dysfunction and oxidative stress have also been put forward as major contributors to the development or maintenance of lipid-induced insulin resistance in muscle. Several animal models with gene deletions in pathways of fatty acid synthesis and storage also show increased metabolic rate, reduced intramuscular lipid storage and improved insulin action when challenged with a high lipid load. SUMMARY: Studies in genetic and dietary obese animal models, genetically modified animals and humans with obesity or type 2 diabetes suggest plausible mechanisms for effects of fatty acids, lipid metabolites, inflammatory pathways and mitochondrial dysfunction on insulin action in muscle. Many of these mechanisms, however, have been demonstrated in situations in which lipid accumulation (obesity) already exists. Whether the initial events leading to muscle insulin resistance are direct effects of fatty acids in muscle or are secondary to lipid accumulation in adipose tissue or liver remains to be clarified.